INCH-POUND

MIL-C-24643/39D
22 November 1994
SUPERSEDING
MIL-C-24643/39C
14 March 1994

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS1S75MU (INCLUDING VARIATION LS1S75MA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

This cable can be manufactured in two variations: LS1S75MU (unarmored), and LS1S75MA (armored).

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor, tin coated, ASTM B 286, size 22-7. Maximum diameter 0.033 inch.

Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.

Third - Cross-linked polyethylene insulation. Nominal wall thickness 0.070 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements.

Fourth - Braided shield of AWG no. 34 tin coated copper. Braid angle of 30 to 35 degrees; minimum coverage of 85 percent.

Fifth - Shield insulation of two polyester tapes, type G in accordance with MIL-I-631, outer tape sealed. An alternate shield insulation may consist of one tape, type G of MIL-I-631, plus an extruded jacket of a suitable transparent material, minimum average thickness of the transparent material 0.003 inch. The standard identification code shall be applied by method 2 on the inner tape.

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Construction (non-watertight) (continued)

Sixth - The eight singles cabled together with a lay not greater than 8 inches around a center filler. Cabling sequence shall be consecutive, starting with no. 1.

Seventh - Binder tape, applied helically with overlap.

Constructive variant for type LS1S75MU

Eighth - Cross-linked polyolefin jacket. Minimum average wall thickness 0.050 inch.

Constructive variant for type LS1S75MA

Eighth - Cross-linked polyolefin jacket. Minimum average wall thickness 0.050 inch.

Ninth - Braided metal armor.

TABLE I. Details.

Military 1/ part no. M24643/39	Size 2/	Number of Conductors	Diameter over completed conductor (inch) (nominal)	Overall diameter (inches)			
				TYPE LS1S75MU		TYPE LS1S75MA	
				min	max	min	max
-01UN	8	8	0.228	0.950	1.030	1.000	1.080

1/ Unarmored. When armor option or double shield is required, see MIL-C-24643 for configuration of part number.

2/ Type LS1S75MU and LS1S75MA.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor Resistance - ohms/1000 feet at 25°C, maximum ...	17.37
Voltage withstand - volts, root mean square, minimum	
Conductor to shield	7000
Shield to shield (component shields only)	500
Insulator resistance - Megohms/1000 feet minimum	
Conductor to shield	500
Component shield to shield	100
* Conductor and shield continuity	No failure

Group A:	Requirements
Visual and dimensional	No failure
* Characteristic impedance - at 1MHz, ohms	75 ± 3

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EXAMINATION AND TESTS (continued):	Requirements
Group B:	
Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When test at 200°C)	50
Attenuation - at 400 MHz, decibel (db)/100 FT maximum	9.0
Cold bending, cable - at 0 ± 2°C, (see table I for mandrel diameter)	No damage
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals - (aged)	
Insulation	
Air Oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged minimum	80
Jacket (cable)	
Air Oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged minimum	60
Hot oil immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged minimum	50
Shrinkage	No failure
Heat distortion - percent of unaged, maximum	30
* Permanence of printing (jacket) - cycles, minimum	125
Armor (type LS1S75MA only) - conformance to material, construction and coverage	No failure
Group D:	
Flame propagation (cable)	No failure

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QUALIFICATION INSPECTION

	Requirements
Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:	
Aging and compatibility (cable)(125 ± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - precent, maximum	
Jacket	2
Fillers	2
Insulation	18
Shield insulation	2
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS: 1000 feet nominal.

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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